

WIRELESS ULTRASONIC RADAR

With TFT Monitor

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The device described is a scaled-down version of a true radar system. It utilises ultrasonic sound waves to detect nearby objects, presenting their precise distance (in cm) and angle (in degrees) on a 10.9cm (4.3-inch) TFT LCD monitor. The TFT LCD features a radar dial-type graphical display with an object scanning animation, creating the appearance of a genuine radar monitor. The device performs real-time object scanning over a 0-180° (half-circle) range. When an object is detected within this range, its distance and angle are displayed on the monitor. Fig. 1 illustrates the three stages of the radar display on the LCD.

As shown in Fig. 1, the TFT LCD provides a radar dial-type half-circle display with angles 15°, 30°, 45°, and so on. It also shows the scanning process animation. When an object is detected, it displays its distance and angle and marks the radar dial with red lines.

As this is a wireless radar system, it consists of two units: the radar transmitter unit, and the radar receiver unit.

The radar transmitter unit continuously scans a half-circle area with a radius of approximately 500cm using ultrasonic sound waves. When it detects any object, it measures its distance and angle and transmits both pieces of data to a remote receiver located at a distance. The transmission range is more than one kilometre in line of sight (LoS). The receiver displays the distance and angle of the object on the TFT LCD and shows the radar dial and scanning process animation.

The transmitter uses an ultrasonic

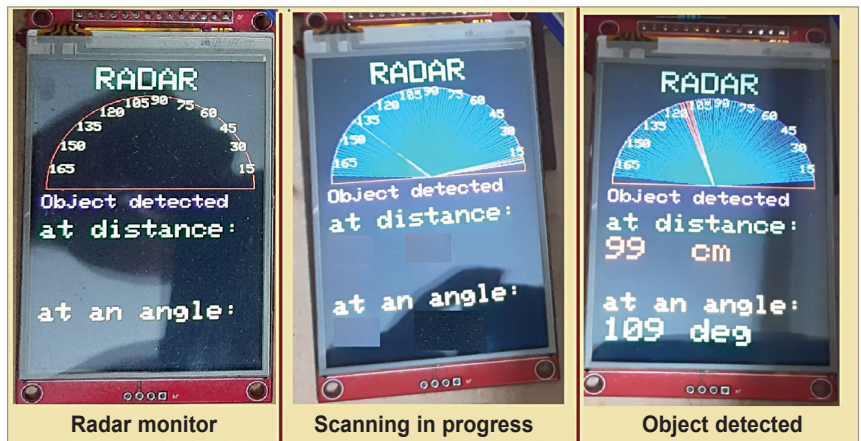


Fig. 1: Three stages of radar on LCD

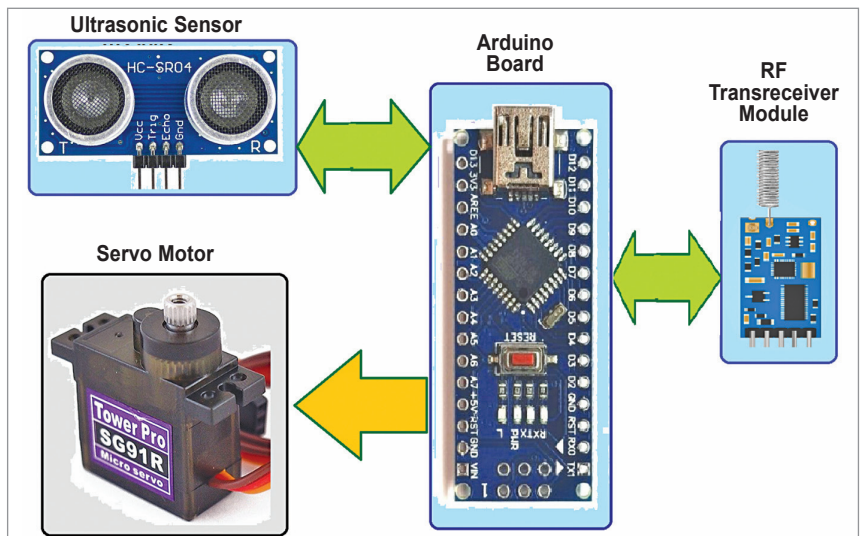


Fig. 2: Block diagram of transmitter

distance measurement (UDM) sensor (HC-SR04), an Arduino Nano board as the microcontroller unit (MCU), and a wireless UART RF transceiver module (HC-12). The receiver also comprises an Arduino board and an HC-12 module, along with a 10.9cm (4.3-inch) SPI TFT LCD with a reso-

lution of 480x320. The components needed for the device are listed in the Bill of Materials table.

Circuit and working

First, let us see how the radar scanner system works through the system block diagram.